

Problem Set 5

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Problem 1

Problem 1.1. Consider a metric of the form

$$ds^2 = dr^2 + f(r)d\Omega_2^2.$$

Assuming the metric is maximally symmetric, find the most general form for $f(r)$. Hint: there are two free parameters corresponding to the scalar curvature and an integration constant, and the integration constant can be removed without loss of generality by shifting $r \rightarrow r + c$.

Solution. Recall a space of dimension 3 is maximally symmetric if and only if the Riemann tensor is of the form

$$R_{\mu\nu\rho\sigma} = \frac{R}{6} (g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}).$$

We first need to compute the Ricci scalar.

```
1 coords = {r, \[Theta], \[Phi]};
2 n = 3;
3 g = DiagonalMatrix[{1, f[r], f[r] Sin\[Theta]^2}];
4 ginv = Inverse[g];
5
6 (* christoffels *)
7 \[CapitalGamma][k_, i_, j_] := 1/2 Sum[ginv[[k, l]] * (D[g[[l, j]], coords[[i]]] +
8   D[g[[i, l]], coords[[j]]] - D[g[[i, j]], coords[[l]]]),
9   {l, 1, n}];
10
11 (* mr riemann ur tensor sucks *)
12 Rtensor[i_, j_, k_, l_] :=
13   D[\[CapitalGamma][i, j, l], coords[[k]]] - D[\[CapitalGamma][i, j, k], coords[[l]]] +
14   Sum[\[CapitalGamma][i, m, k] \[CapitalGamma][m, j, l] - \[CapitalGamma][i, m, l]
15     \[CapitalGamma][m, j, k], {m, 1, n}];
16
17 (* ricci *)
18 RicciTensor[i_, j_] := Sum[Rtensor[m, i, m, j], {m, 1, n}];
19 RicciScalar = Simplify[
20   Sum[ginv[[ii, jj]] RicciTensor[ii, jj], {ii, 1, n}, {jj, 1, n}]
21 ]
```

This code yields

$$R = \frac{f'(r)^2 + 4f(r)(1 - f''(r))}{2f(r)^2}.$$

This gives R in terms of derivatives of f . We can solve for f in terms of R by solving the second order ode. This is in fact an ODE because we can take R to be constant in a maximally symmetric space since curvature is constant. The general solution is a mess, so we modify the equation to make it easier for Mathematica to solve. If we set $f(r) = a(r)^2$ then we obtain

$$R = \frac{\left(\frac{d}{dr}a(r)^2\right)^2 + 4a(r)^2(1 - (a(r)^2)'')}{2a(r)^4} = \frac{4a(r)^2a'(r)^2 + 4a(r)^2(1 - 2a(r)a''(r) - 2a'(r)^2)}{2a(r)^4}$$

$$= \frac{2a'(r)^2 + 2 - 4a(r)a''(r) - 4a'(r)^2}{a(r)^2} = \frac{2 - 2a'(r)^2 - 4a(r)a''(r)}{a(r)^2}.$$

We then obtain

$$Ra^2 = 2 - 2(a')^2 - 4aa''.$$

We also perform a change of variables $u(a) = \left(\frac{da}{dr}\right)^2$. This just kind of made Mathematica solve the integral better. We can take $a(0) = 0$ since at the origin there is no θ -dependence, so the $d\Omega_2^2$ term in the metric should vanish. We can also take $a'(0) = 1$ since near the origin curvature becomes negligible from the observer's frame, and flat space has $ds^2 = dr^2 + r^2 d\Omega_2^2$. This yields $a \rightarrow r$, and thus $a' \rightarrow 1$.

```

1 odeA = Simplify[RicciScalar == R /. f -> (a[#]^2 &)];
2 reducedODE = Simplify[
3   odeA /. {a'[r] -> u'[a]/2, a'[r] -> Sqrt[u[a]], a[r] -> a}
4 ];
5 (* boundary condition *)
6 solU = DSolveValue[{reducedODE, u[0] == 1}, u[a], a];
7 SolveGeneral[assumption_] := Module[{integral, invSol, aSol},
8   (* find r as a function of a *)
9   integral = Simplify[
10    Integrate[1/Sqrt[solU], a, Assumptions -> {a > 0, assumption}],
11   assumption
12 ];
13 (* recover a as a function of r *)
14 invSol = Solve[r == integral, a, Assumptions -> {r > 0, assumption}];
15
16 aSol = a /. Last[invSol]
17 (* recover f = a^2 *)
18 Simplify[aSol^2, assumption]
19 ];
20 solPos = SolveGeneral[R > 0];
21 Print[solPos];
22 solZero = SolveGeneral[R == 0];
23 Print[solZero];
24 solNeg = SolveGeneral[R < 0];
25 Print[solNeg];

```

This outputs

$$\begin{aligned}
R > 0 &\implies f(r) = \frac{6 \sin\left(\frac{r\sqrt{R}}{\sqrt{6}}\right)^2}{R} \\
R = 0 &\implies f(r) = r^2 \\
R < 0 &\implies f(r) = -\frac{6 \sinh\left(\frac{r\sqrt{-R}}{\sqrt{6}}\right)^2}{R}.
\end{aligned}$$

Problem 2

Problem 2.1. Show that when someone falls across the Horizon of a Schwarzschild black hole, the slowest they can approach the center $r = 0$ is

$$\left|\frac{dr}{d\tau}\right| \geq \sqrt{\frac{2GM}{r} - 1},$$

for $d\tau$ the proper time.

Solution. We start with the Schwarzschild metric

$$ds^2 = - \left(1 - \frac{R_S}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{R_S}{r}} + r^2 d\Omega^2.$$

Let $d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu$. Take $r < R_S$, making $\frac{R_S}{r} > \frac{R_S}{R_S} = 1$. Thus $1 - \frac{R_S}{r} < 0$. We can factor out a negative sign, yielding

$$d\tau^2 = - \left(\left(\frac{R_S}{r} - 1 \right) dt^2 - \frac{dr^2}{\frac{R_S}{r} - 1} + r^2 d\Omega^2 \right) = - \left(\frac{R_S}{r} - 1 \right) dt^2 + \frac{dr^2}{\frac{R_S}{r} - 1} - r^2 d\Omega^2.$$

We start by maximizing $d\tau/dr$, and recover the minimum of $dr/d\tau$ using the fact that they are inverses.

To maximize $d\tau^2$, we note that the dt and $d\Omega$ terms are negative, so the largest they can be is 0. Assuming they vanish, the maximum of $d\tau^2$ is

$$d\tau^2 \leq \frac{dr^2}{\frac{R_S}{r} - 1} \implies \frac{d\tau^2}{dr^2} \leq \frac{1}{\frac{R_S}{r} - 1} = \frac{1}{\frac{R_S - r}{r}} = \frac{r}{R_S - r}.$$

Inverting the fraction yields

$$\frac{dr^2}{d\tau^2} \geq \frac{R_S - r}{r} = \frac{R_S}{r} - 1.$$

Since $R_S = 2GM$,

$$\left| \frac{dr}{d\tau} \right| \geq \sqrt{\frac{R_S}{r} - 1} = \sqrt{\frac{2GM}{r} - 1}.$$

Problem 2.2. Show if they go as slow as possible, then they go from the horizon $r = 2GM$ to the singularity $r = 0$ in a proper time of $\Delta t = \frac{\pi}{2} R_S$. Hint: use the identity $\int_0^1 \frac{1}{\sqrt{x^{-1}-1}} dx = \frac{\pi}{2}$; restore the factors of c by dimensional analysis.

Solution. We continue using $R_S = 2GM$. Since the person is falling from the horizon to the singularity, r is decreasing, so we write it with a negative sign:

$$d\tau = - \frac{dr}{\sqrt{\frac{R_S}{r} - 1}}.$$

Integrating gives

$$\Delta\tau = \int_{R_S}^0 - \frac{1}{\sqrt{\frac{R_S}{r} - 1}} dr = \int_0^{R_S} \frac{1}{\sqrt{\frac{R_S}{r} - 1}} dr.$$

In order to use the given identity, we substitute r for a dimensionless quantity $R_S x = r$, $R_S dx = dr$. We then have

$$\Delta\tau = \int_0^1 \frac{R_S}{\sqrt{x^{-1}-1}} dx = \frac{\pi}{2} R_S.$$

The hint also tells us to restore factors of c by dimensional analysis, so note that the Schwarzschild radius has units of length and proper time has units of time. We thus divide by velocity c to obtain $\Delta\tau = \frac{\pi R_S}{2c}$.

Problem 2.3. *The largest black hole discovered so far has a Schwarzschild radius of 9.75×10^{13} meters. Calculate the maximum $\Delta\tau$ using the formula from problem 2.2.*

Solution. The meter is defined such that

$$c = 299,792,458.$$

We have

$$\Delta\tau = \frac{\pi}{2} \frac{9.75 \times 10^{13}}{299,792,458} \approx 510,862 \text{ seconds.}$$

According to Google that's almost 6 days.